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COURSE NAME : SOFTWARE

ENGINEERING

CO3-ASSESSMENT TOOL - 01

Title: AI-Powered Personalized Nutrition and Health Recommendation Platform

1. Problem Analysis

Problem Statement

Generic diet recommendations do not consider an individual's age, weight, lifestyle, health condition, physical activity, dietary preferences, and fitness goals. This can result in unsuitable nutrition plans and poor health outcomes.

The proposed AI-Powered Personalized Nutrition and Health Recommendation Platform collects user health information and analyzes it using AI-based recommendation techniques. The system generates personalized meal plans, monitors nutritional intake, tracks health progress, and provides continuous recommendations.

Objectives

1. Collect personal health information.
2. Analyze nutritional requirements using AI.
3. Generate personalized meal plans.
4. Monitor daily dietary intake.
5. Integrate wearable health information.
6. Provide personalized health recommendations.
7. Track nutrition and health progress.
8. Encourage healthy lifestyle adoption.

Functional Requirements

Requirement	Description
User Registration	Users can create an account.
Login	Users can securely access their account.
Health Data	Users enter age, weight, height, activity level, etc.

Requirement	Description
Nutrition Analysis	System analyzes the user's nutritional requirements.
AI Recommendation	AI generates personalized recommendations.
Meal Planning	System generates suitable meal plans.
Nutrition Tracking	Users can record consumed food.
Progress Monitoring	System displays nutrition/health progress.
Wearable Integration	Health information can be obtained from wearable devices.
Notifications	System provides reminders and recommendations.

Expected Outcomes

- Personalized nutrition guidance
- Improved dietary habits
- Better health management
- Continuous nutrition monitoring
- Increased user engagement
- Improved accessibility to nutrition guidance
- Support for healthy lifestyle adoption

2. Project Structure Design

Use a simple structure so that your Git repository is easy to understand.

```

AI-Nutrition-Platform/
|
├── backend/
│   ├── app.py
│   ├── requirements.txt
│   ├── models/
│   │   └── recommendation.py
│   ├── routes/
│   │   ├── user.py
│   │   └── nutrition.py
│   └── database/
│       └── database.py
└── frontend/
    ├── index.html
    ├── login.html
    ├── dashboard.html
    └── style.css

```

```
├── tests/
│   ├── test_user.py
│   └── test_nutrition.py
├── Dockerfile
├── docker-compose.yml
├── .dockerignore
├── .gitignore
├── README.md
└── requirements.txt
```

Purpose of Major Folders

- **backend/**
Contains the server-side application and AI recommendation logic.
- **frontend/**
Contains the user interface of the nutrition platform.
- **models/**
Contains AI/recommendation-related logic.
- **routes/**
Contains APIs for users and nutrition-related operations.
- **database/**
Contains database connection and data management code.
- **tests/**
Contains test programs used to verify application functionality.
- **Dockerfile**
Defines how the application Docker image is created.
- **docker-compose.yml**
Defines and manages multiple application services.
- **.gitignore**
Specifies files that Git should not track.
- **.dockerignore**
Prevents unnecessary files from being copied into the Docker image.
- **README.md**
Provides project information, installation instructions, and execution steps.

3. Git Repository Initialization

Git is used to maintain different versions of the project and allow multiple developers to work collaboratively.

Step 1: Create Project Folder

Create:

```
AI-Nutrition-Platform
```

Open the folder in VS Code.

Step 2: Open Terminal

Run:

```
git init
```

This creates a Git repository.

Step 3: Check Git Status

```
git status
```

Step 4: Add Project Files

```
git add .
```

Step 5: Create First Commit

```
git commit -m "Initial project setup"
```

Step 6: Connect Remote Repository

Create a repository on GitHub named:
AI-Nutrition-Platform

Then:

```
git remote add origin YOUR_GITHUB_REPOSITORY_URL
```

Finally:

- `git branch -M main`
- `git push -u origin main`

Important Git Files

File	Purpose
.git/	Stores Git repository information and history
.gitignore	Prevents unwanted files from being committed
README.md	Project documentation
.gitconfig	Git configuration on the developer system

Example .gitignore:

```
__pycache__/  
*.pyc  
.env  
venv/  
.venv/  
node_modules/
```

4. Benefits of Git and Docker

Area	Git	Docker
Collaboration	Multiple developers can work together	Same environment for all developers

Area	Git	Docker
Version Management	Maintains complete code history	Maintains consistent application environments
Portability	Code can be shared through repository	Application can run on different systems
Deployment	Code can be synchronized	Containers simplify deployment
Scalability	Supports parallel development	Multiple containers can be deployed
Maintainability	Tracks changes	Isolates application dependencies
Reliability	Allows rollback	Provides consistent execution environment

Git Benefits

Git provides:

- Version control
- Branch management
- Collaboration
- Change tracking
- Rollback capability
- Conflict management
- Complete project history

Docker Benefits

Docker provides:

- Environment consistency
- Dependency isolation
- Easy deployment
- Portability
- Faster setup
- Container-based scalability
- Easier testing

5. Challenges and Improvements

Challenges

1. Git Merge Conflicts

Different developers may modify the same files.

Solution:

Use feature branches and regularly synchronize with the development branch.

2. Dependency Conflicts

Different Python libraries may require different versions.

Solution:

Use:

requirements.txt

and Docker.

3. Database Connectivity

The backend may initially fail to connect to PostgreSQL.

Solution:

Use the Docker Compose service name:

database

instead of localhost.

4. Container Configuration

Incorrect ports or environment variables can prevent the application from starting.

Solution:

Maintain configuration using environment variables.

5. Large AI Models

AI models can make Docker images very large.

Future improvement:

Use lightweight models or separate the AI recommendation service into its own container.

Future Enhancements

The project can be improved by adding:

1. Advanced AI nutrition prediction.
2. Real-time wearable integration.

3. Mobile application.
4. Cloud deployment.
5. Authentication using JWT.
6. Automated CI/CD using GitHub Actions.
7. Kubernetes-based container orchestration.
8. AI chatbot for nutrition queries.
9. Personalized exercise recommendations.
10. Real-time health alerts.

Git + Docker Workflow Diagram

